

Flight Delay Prediction

Exploratory Data Analysis Summary

Key Findings & Business Insights

January 16, 2026

Executive Summary

This exploratory data analysis examines 148,052 flight records from 2017-2018 to identify key patterns and predictors of flight delays. Our analysis reveals significant temporal, geographic, and operational patterns that can inform predictive modeling and business decisions.

Dataset Overview:

Metric	Value
Total Flights	148,052
On-Time Flights	126,740 (85.6%)
Delayed Flights (>15 min)	21,312 (14.4%)
Time Period	2017-2018

1. Temporal Patterns

1.1 Time of Day Analysis

Time of day is the strongest predictor of flight delays. Our analysis reveals dramatic variations in delay rates throughout the day:



KEY FINDING: Late night flights (0-3 AM) show 100% delay rates, while early morning flights (5-7 AM) have the lowest delay rates at 2.9-6.4%.

Hourly Delay Rate Breakdown:

- Early Morning (5-7 AM): 2.9-6.4% delay rate - BEST TIME TO FLY
- Morning (8-11 AM): 7.6-9.2% delay rate - Good reliability
- Afternoon (12-5 PM): 10.9-15.2% delay rate - Moderate risk
- Evening (6-10 PM): 16.8-22.3% delay rate - High risk
- Late Night (0-3 AM): 65-100% delay rate - AVOID

Business Implication: Delays compound throughout the day. A single morning delay cascades through afternoon and evening flights as aircraft rotate through multiple routes.

1.2 Day of Week Patterns

Day of week significantly impacts delay probability:

Day	Delay Rate	Risk Level
Monday	17.6%	HIGH
Friday	16.0%	HIGH
Tuesday	13.2%	LOW
Wednesday	12.9%	LOW
Saturday	12.4%	LOW

Insight: Monday has 42% higher delay rates than Tuesday/Wednesday. This is likely due to aircraft repositioning over the weekend and higher business travel volume on Mondays.

1.3 Seasonal Patterns

Summer months show significantly higher delay rates:

- Summer Peak (June-July): 21.0-21.2% delay rate
- November (Best): 6.5% delay rate (lowest of the year)
- Winter/Spring: 12-14% average delay rate



BUSINESS RECOMMENDATION: Increase buffer times for summer travel and implement dynamic delay warnings during June-July peak season.

2. Geographic Patterns

2.1 Highest-Risk Origin Airports

The following airports show the highest delay rates for departing flights (minimum 100 flights analyzed):

Airport	City	Delay Rate
PIA	Peoria, IL	25.0%
MGM	Montgomery, AL	21.7%
MOB	Mobile, AL	21.2%

2.2 Highest-Risk Destination Airports

Airport	City	Delay Rate
EWB	Newark, NJ	28.4%
JFK	New York, NY	22.8%
SFO	San Francisco, CA	22.0%
BOS	Boston, MA	20.8%

Key Observation: Major hub airports (EWB, JFK, SFO, BOS) show higher delay rates due to congestion and complex operations. Newark (EWB) has nearly double the national average delay rate.

3. Carrier Performance Analysis

Significant variation exists across carriers, with delay rates ranging from 10.7% to 22.2%:

3.1 Best Performing Carriers

Carrier	Delay Rate	Flight Volume
AS (Alaska)	10.7%	262
YX (Republic)	11.4%	676
DL (Delta)	12.0%	95,931 (largest)

3.2 Highest Delay Carriers

Carrier	Delay Rate	Flight Volume
B6 (JetBlue)	22.2%	1,283
OO (SkyWest)	21.1%	7,387
WN (Southwest)	20.1%	16,370

 **INSIGHT:** Delta (DL) combines high volume (95,931 flights) with low delay rates (12%), indicating operational excellence. This makes it statistically the most reliable carrier in the dataset.

4. Distance & Route Analysis

Flight distance shows a non-linear relationship with delays:

Distance Group	Range (miles)	Delay Rate
Group 8	1,750-1,999	16.3%
Groups 3-4	500-1,249	15.1%
Group 1-2	< 500	13.5-13.9%
Group 6	1,250-1,499	9.9%

Key Finding: Short-haul flights (<500 miles) and very long flights (>2,000 miles) show lower delay rates. Mid-range flights (1,750-2,000 miles) have the highest delay probability at 16.3%.

5. Feature Correlation Analysis

Correlation analysis reveals which features have predictive power for delays:

5.1 Strongest Correlations with Delay

- HOUR (Time of Day): +0.15 correlation - Strongest predictor
- DISTANCE: +0.01 correlation - Weak predictor
- YEAR, QUARTER, MONTH: -0.02 to -0.03 - Minimal correlation
- DAY_OF_WEEK: -0.03 correlation - Weak negative (weekend effect)

STATISTICAL INSIGHT: HOUR has 15x stronger correlation with delays than any other numeric feature. This validates our temporal analysis showing time-of-day as the primary delay driver.

6. Business Recommendations

6.1 For Travelers

- **Book early morning flights (5-7 AM) for 70% lower delay risk**
- Avoid Monday and Friday travel when possible (33% higher delays)
- Travel in November for lowest seasonal delays (6.5%)
- Choose Delta (DL) or Alaska (AS) for best on-time performance
- Add extra buffer time for Newark (EWR), JFK, and SFO destinations

6.2 For Travel App Development

- **Implement time-of-day risk scoring (highest priority feature)**
- Display carrier delay rates in booking interface
- Provide 'Best Time to Fly' recommendations for each route
- Alert users 24-48 hours before high-risk flights
- Offer alternative flight suggestions with lower delay probability
- Create seasonal travel guides (avoid June-July peak)

6.3 For Airlines

- Focus operational improvements on afternoon/evening time slots
- Increase buffer times for Newark, JFK, and SFO operations
- Optimize Monday schedules to handle business travel surge
- Pre-position aircraft more effectively on weekends
- Add capacity during June-July peak delay season

7. Implications for ML Model Development

7.1 Key Predictive Features Identified

Feature Type	Features	Priority
Temporal	Hour, Day of Week, Month, Season	CRITICAL
Geographic	Origin, Destination, Hub Status	HIGH
Operational	Carrier, Route Popularity	HIGH
Distance	Distance Group, Distance Category	MEDIUM

7.2 Class Imbalance Challenges

The dataset exhibits significant class imbalance (85.6% on-time vs 14.4% delayed), requiring specialized handling:

- **Use SMOTE or class weighting to balance training data**
- Optimize for F1-score and Recall rather than Accuracy
- Consider ensemble methods (XGBoost, CatBoost, Random Forest)
- Implement threshold optimization for production deployment
- Focus on minimizing false negatives (missed delays)

7.3 Expected Model Performance

Based on EDA findings and class distribution, we expect:

- Target F1-Score: 0.40-0.50 (good for imbalanced data)
- Target Recall: 0.50-0.65 (catch 50-65% of delays)
- Target Precision: 0.35-0.45 (minimize false alarms)

- Accuracy alone will be misleading (>80% by predicting all on-time)

8. Conclusion & Next Steps

8.1 Key Takeaways

- **Time of day is the strongest delay predictor**
- **Monday and Friday show 30-40% higher delays**
- **Summer travel (June-July) has 3x higher delays than November**
- Newark (EWR) has 2x the national average delay rate
- Carrier choice matters: 10.7% (Alaska) vs 22.2% (JetBlue)
- Class imbalance (85/15 split) requires specialized ML techniques

8.2 Next Steps

- **Develop predictive models using XGBoost, CatBoost, and ensemble methods**
- Implement proper class balancing techniques (SMOTE, class weights)
- Engineer interaction features (hour × day, carrier × route)
- Optimize decision threshold for business objectives
- Integrate weather data for improved predictions
- Deploy A/B testing framework before full rollout

This analysis provides a strong foundation for building an accurate, production-ready flight delay prediction system that can deliver significant business value through improved user experience and operational efficiency.

--- End of Report ---